

Twin Flame VAJRA

Finite Ecological Closure, Registered Semantics, and Reciprocal Adjudication

Technical and Academic Companion to TF-NE10 — Finite Ecological Closure Demonstrator

Review Draft v0.1 · source-grounded manuscript master

This paper is designed to stand alone and to function as the formal companion to the TF-NE10 interactive demonstrator. Experimental results, definitions, demonstrations, interpretations, open questions, and failed gates are deliberately typed separately.

Scope: finite constructive closure at the frozen fixtures. No claim of infinite recursion, universal scale invariance, metaphysical identity, general intelligence, or truth-from-survival is made.

Abstract

This paper consolidates a finite experimental and formal chain developed in the MetaLabe / Twin Flame Crucible program. The empirical component establishes, within frozen computational fixtures, that ecological consequence can produce reciprocal niche-specific specialization; that the resulting organization can remain functionally consequential through a declared compactification interface; that selection at an intermediate organizational grain can materially alter adaptation observed one grain above; and that environmental residence time strongly modulates tracking without any tested rate satisfying the prospectively defined full-TRACKED gate. The formal component specifies a receipt-governed adjudication system in which claims are registered by propositional content, attribution, domain, quantifier structure, grain, conditions, modal/epistemic status, and inferential strength. It then defines semantic substitution, guarded reflexive restoration, side-index dependence, and a reciprocal relational state called VAJRA. Symbolic terms including Twin Flame, Solve et Coagula, As Above, So Below, and VAJRA are treated as registered structural vocabulary: neither authorities nor disqualifiers, but names whose admissibility depends on precise referents and the same evidentiary standard applied to conventional terminology. The resulting closure is finite and scoped. It establishes neither an infinite recursive ladder nor a terminal grain. It instead supplies a reproducible boundary between what the frozen evidence earns, what it fails to earn, and what remains OPEN.

Plain-language abstract

The experiments show that organization can live in relationships, not only in isolated parts. When consequences differ across ecological roles, initially identical adaptive populations can specialize in complementary ways. That organization can still matter after the system is compressed into a higher-level representation. At a second finite level, removing intermediate selection can materially reduce higher-level recovery. Slower environmental switching improves tracking, but none of the tested rates crossed the locked full-tracking threshold. The formal result is equally important: evidence must constrain positive and negative claims by the same rule. A failure to observe something at a finite tested scope does not by itself license a claim that the thing is impossible everywhere. VAJRA names the state of a reciprocal pairing that remains distinct, revisable, and mutually challengeable while preserving unresolved residue as OPEN rather than forcing agreement or severance.

Claim typing used throughout

Type	Meaning
EXPERIMENTAL RESULT	Frozen measured outcome tied to a prospective lock, runner/results artifact, gate, and scope.
DERIVED / THEOREM	Formal consequence of declared definitions, axioms, or premises.
DEFINITION / CLASSIFICATION	A stipulated operational vocabulary or classifier; not an empirical receipt by itself.
DEMONSTRATION	An executable or worked illustration of a definition/classifier; not evidence about an external target unless separately bound to one.
FAILED / BOUNDED	A prospectively declared stronger gate that was not met, or a topology/condition where materiality was not earned.
OPEN	Not licensed for promotion or nullification by available

	receipts at the requested scope.
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1. Problem statement: closure without asymmetry

The project asks a narrower question than whether any favored worldview is ultimately true. It asks what can be promoted when an adjudicator has declared a receipt rule, and whether that rule is applied symmetrically. The Double-Edged Receipt Standard separates truth, warranted promotion at a declared scope, and consistency of adjudication. Its constitutional rule is that no unreceipted load-bearing edge may be silently promoted.

Source receipt: Double_Edged_Receipt_Standard working paper: claim scope, receipt jurisdiction, adequacy, load-bearing dependency, receipt-governed adjudicator, R0, and Double-Edged Receipt Theorem.

1.1 Double-Edged Receipt Theorem

Let C be a claim; $\sigma(C)$ its declared scope/registration; R a receipt; $J(R)$ the jurisdiction actually supported by R ; $\text{Adeq}(R, C)$ receipt adequacy; $\text{Dep}(C)$ load-bearing dependencies; and S a receipt-governed adjudicator.

R0: no non-axiomatic load-bearing dependency may be silently promoted without an adequate receipt or explicit OPEN/conditional status.

If S requires $\text{Adeq}(R, G)$ before promoting G , S cannot consistently promote $\neg G$ (or a rejection entailing $\neg G$) through a load-bearing non-axiomatic premise B for which no adequate receipt is supplied.

Proof sketch. If the promoted negative conclusion depends load-bearingly on B , then by the definition of a receipt-governed adjudicator and R0, B must be adequately receipted, explicitly axiomatic, or left conditional/OPEN. Promoting the negative while B lacks those statuses violates the same rule invoked against G . The theorem does not prove G ; it proves that skepticism cannot receive an undeclared exemption from the receipt rule.

1.2 Scope is not a scalar

For TF-NE10, the registration is expanded to an extensible coordinate tuple:

$$\sigma(C) = \langle P, A, D, Q, G, K, M, I \rangle$$

Coordinate	Meaning
P	Propositional content / predicate.
A	Attribution or speaker commitment.
D	Domain.
Q	Quantifier structure and order.
G	Grain / resolution.
K	Conditions and context.
M	Modal or epistemic status.
I	Inferential strength.

The adjudication standard S is not a proposition coordinate. The registry is explicitly extensible: additional coordinates may be introduced when they are truth-material for a declared problem. Declaring a transformation makes it visible; it does not by itself license transfer of a verdict.

2. Registered semantics and semantic substitution

A semantic substitution occurs when an adjudication directed at registered claim C is transferred to a materially different claim C' without the truth-material transformation being made explicit. This is not proposed as a new fallacy. It is a registration and classification scheme over recognized errors such as equivocation, ignoratio elenchi, straw man, illicit generalization, quantifier shift, existential import, secundum quid, and argument from ignorance.

Witness $W = (C, C', T)$, where T identifies the changed coordinate(s) such that $\sigma(C) \neq \sigma(C')$.

Restoration test: restore the changed coordinates of C' to those of C . If the alleged contradiction or inference still survives, the claimed substitution witness fails. If restoration removes it, the witness succeeds for that transformation.

This guard prevents the classifier from becoming self-sealing. An objection that preserves the registered coordinates may genuinely defeat or refine the primitive.

2.1 Declaration is not transfer

Visible transformation $\neq$ licensed inference.

The classifier therefore asks three separate questions: what changed; was the transformation declared; and does an applicable inference, monotonicity, closure, or other transfer rule license carrying the adjudication across that change?

2.2 Corrected classification map

Structural change	Classical classification / condition
Same surface term, changed meaning	Equivocation.
C answered by establishing or attacking C'	Ignoratio elenchi.
$C \rightarrow C'$, C' attributed to opponent, then attacked	Straw man; attribution A discriminates it from generic irrelevance.
$\exists x P(x) \rightarrow \forall x P(x)$	Hasty / illicit generalization.
$\forall x \exists y R(x,y) \rightarrow \exists y \forall x R(x,y)$	Quantifier shift: order in Q changes.
$\forall x P(x) \rightarrow \exists x P(x)$ over possibly empty D	Existential fallacy / unwarranted existential import.
$P(x K) \rightarrow P(x)$	Secundum quid: qualified statement taken absolutely.
not-established(P) $\rightarrow$ $\neg P$, or not-established($\neg P$) $\rightarrow P$	Argument from ignorance via M , unless a declared justified closure/completeness license applies.
$S(C1) \neq S(C2)$ absent relevant supported distinction	Special pleading / adjudicator-level asymmetry, not proposition substitution.

2.3 Flagship NE9 worked example

PLAIN CLAIM	No tested residence time met the frozen full-TRACKED gate. This does not mean tracking is impossible at every possible rate.
FORMAL REGISTRATION	$C = \neg\exists\tau \in D_{\text{tested}} : \text{TRACKED}(\tau)$. Unsupported strengthening: $C' = \Box \neg\exists\tau \in D_{\text{possible}} : \text{TRACKED}(\tau)$. Changed coordinates: D (tested→possible) and M (scoped actual negative→necessary negative). P is held fixed.
RECEIPT / INFERENCE	Restoring D and M in C' returns C exactly. A separate TRACKED→Closure replacement would be an additional P-coordinate substitution and requires its own registered bridge. Thus finite non-achievement cannot silently become necessary impossibility.
BOUNDARY	OPEN remains appropriate for slower rates and broader domains. No claim is made that full tracking must eventually occur.

3. Mechanical bias as side-index dependence

Bias is used here mechanically, not motivationally. Let ι denote a side index such as self/other, familiar/unfamiliar, proponent/challenger, or another declared relational position. If claim registration, receipts, and legitimate conditions are held fixed, an adjudication rule that changes solely with an unlicensed ι depends on that index.

Invariant form: $S(C) = f(\sigma(C), R)$.

Side-dependent form: $S(C) = f(\sigma(C), R, \iota)$.

Exchange test: if ι is not independently relevant, swapping sides while holding $\sigma(C)$, R , and the admissible rule fixed should leave the burden unchanged.

This detects dependence, not motive. A structural asymmetry can be observed without asserting why an adjudicator produced it. Relevant asymmetries remain admissible when the distinguishing coordinate is declared and adequately supported.

4. Experimental chain

The empirical chain is deliberately finite. TF-NE4 establishes reciprocal ecological specialization; TF-NE5 tests transport through a declared recursive lift and post-shift re-expression; TF-H1 supplies an important bounded failure for a stronger offspring-heredity claim; TF-NE9 tests a second finite recursive rung and residence-time dependence. TF-NE10 adds no new scientific gate; it synthesizes these frozen receipts.

4.1 TF-NE4 — reciprocal niche specialization

PLAIN CLAIM	Initially byte-identical heritable neural populations can develop reciprocal niche-specific specialization when
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	niche consequence controls reproductive opportunity.
FORMAL REGISTRATION	For paired populations derived from identical founders, specialization interaction Δ is evaluated across N1/N2 batteries and three topology strata against selection-disconnected controls. N2 additionally requires cue→action relational organization rather than a single global action bias.
RECEIPT / INFERENCE	Frozen adjudication PASS in all six battery×topology strata; 24/24 paired seeds positive in each. An independent reproduction reported byte-identical exact results, internal results hash match, and founder_match=true in all 144 rows.
BOUNDARY	Finite fixture-bounded result only; no universal convergence, general intelligence, unlimited recursion, or epistemic truth from survival.

Battery / topology	Positive seeds	Median Δ	Null median	Δ - null
N1 hand-minimal	24/24	3.7455	0.0312	3.6429
N1 random-sparse	24/24	3.7812	-0.1652	4.0000
N1 worm-like	24/24	3.7946	-0.1920	4.0402
N2 hand-minimal	24/24	3.4375	0.0156	3.3594
N2 random-sparse	24/24	3.3125	0	3.2031
N2 worm-like	24/24	3.2656	0	3.2812

Source receipt: TF_NE4_PROOF_IN_PUDDING_v0_3_FROZEN_AUDIT.html. Exact results SHA-256

2f5f4c5e41e14fa98dec2c5bb8c7c2ce18549047a6180e5fb8d152de2c1d5f65; internal results_hash

302a1726539529e4c9605e00ae5a8e7a5dfe99acad50a6882c255e9812f2f471. Audit notes distinguish ordinary raw-file SHA-256 from the frozen helper's Buffer-JSON hash convention.

4.2 TF-NE5 — specialization transport across recursive lift

PLAIN CLAIM	Niche-specific organization remained functionally consequential after compactification; fresh lower-population re-expression produced consistent post-shift adaptation unavailable to frozen copying.
FORMAL REGISTRATION	Phase T tests functional visibility after compactification. Phase R compares fresh re-expression with frozen copying after an opposed higher-grain regime shift. The stronger locked materiality gate requires median recovery ≥ 0.50 and median advantage ≥ 0.50 .
RECEIPT / INFERENCE	Phase T passed all three topology strata. Phase R was positive in 24/24 seeds per stratum, while frozen-copy median gain was zero, but observed recovery (~0.10–0.12) and advantage (~0.19–0.20) remained below the

	stronger locked thresholds.
BOUNDARY	The stronger proposition “re-expression preserved material higher-grain adaptive capacity” is not promoted at TF-NE5 scope. Compactification is an interface/provenance operation, not identity between grains.

Topology	Positive gain	Median gain	Recovery	Advantage vs frozen	Gate
hand-minimal	24/24	0.190430	0.098287	0.190430	BELOW MATERIALITY
random-sparse	24/24	0.202148	0.116683	0.202148	BELOW MATERIALITY
worm-like-relay	24/24	0.198242	0.108964	0.198242	BELOW MATERIALITY

Source receipt: TF_NE5_RECURSIVE_LIFT_v0_1_FROZEN.html. Prospective lock
d0da60c79ee784188c18b94aeb7a911c6221d22b86cc93e25a4fd5babfa7a53b; runner
4186ab96f8cd019860b80e5f69f049f921e37a73e8cbe7f4f6899cf15f121beb; exact results
4a52a0fb67a4e1481a94c7eb18c947de2d5caa654be2332446fc015868cbe682; internal results
8379a28732591fbdc9872b23ca5716cd5be02e1e32d604bfc5d4ca024ec30ad0; bound source subset
5097022f3ed706ec838f8afe5895f59480abb7f6fa4076ff7b8254b8045225597.

4.3 TF-H1 — bounded offspring / third-state heredity result

PLAIN CLAIM	The pair-specific provenance signal was real in the frozen fixture, but the stronger relation-conditioned offspring-heredity materiality claim was not earned.
FORMAL REGISTRATION	TRUE_RELATION_INHERITANCE is compared with no-relation recombination, independent-parent history, static pair summary, and shuffled relation under familiar and mirrored contexts. Locked materiality over strongest nonrelational control required ≥ 0.03 .
RECEIPT / INFERENCE	TRUE_RELATION exceeded the strongest nonrelational control in 16/16 seeds in both contexts, but grand-mean margins were only +0.004375 familiar and +0.004614 mirrored. By contrast, TRUE_RELATION exceeded SHUFFLED_RELATION by +0.126611 and +0.127469, satisfying provenance specificity.
BOUNDARY	Classification remains NO_OR_MIXED_RELATION_CONDITIONED_OFFSPRING_HEREDITY_AT_TF_H1_SCOPE. Do not promote this to narrow offspring heredity.

Arm	Familiar	Mirrored
NO_RELATION_RECOMBINATION	0.500113	0.500247

INDEPENDENT_PARENT_HISTORY	0.698566	0.697635
STATIC_PAIR_SUMMARY	0.745689	0.744774
SHUFFLED_RELATION	0.623454	0.621920
TRUE_RELATION_INHERITANCE	0.750065	0.749388

Source receipt: MetaLabe_Twin_Flame_Crucible_R0_1_FULL_PRERELEASE.html / TF-H1 frozen findings: seeds 61001–61016, 256 parent pairs/seed, 16 neutral binary traits, 48 training rounds/context, 32 offspring evaluation rounds/context; exact global inversion max absolute difference 0.0.

4.4 TF-NE9 — second rung and rate dependence

PLAIN CLAIM	Intermediate selection remains causally consequential at a second finite recursive rung in two of three topology strata; environmental residence time strongly affects tracking, but no tested rate reaches the full TRACKED gate.
FORMAL REGISTRATION	Part A ablates intermediate selection and requires ≥ 0.10 median advantage in at least two topology strata, ≥ 0.50 full recovery, and $\geq 20/24$ positive seeds. Part B sweeps $\tau \in \{1, 2, 4, 8, 16\}$; TRACKED requires alignment ≥ 0.50 plus $\geq 20/24$ positive seeds.
RECEIPT / INFERENCE	Part A: random-sparse advantage 0.216 and worm-like 0.172 are material; hand-minimal 0.001 is not. Part B: alignment rises monotonically with τ in all three topologies, endpoint recovery rises strongly, but alignment remains below 0.50 at every tested τ .
BOUNDARY	Finite second-rung result only. No universal recursion law, infinite ladder, scale invariance, unique timescale, or impossibility of tracking at slower rates is earned.

Topology	Full	No intermediate	Advantage	Positive
hand-minimal	0.847	0.845	0.001	24/24
random-sparse	0.859	0.644	0.216	24/24
worm-like-relay	0.646	0.474	0.172	24/24

Topology	τ	Alignment	Endpoint recovery	Positive	Status
hand	1	0.080	0.000	20/24	UNTRACKED
hand	2	0.124	0.023	22/24	PARTIAL
hand	4	0.169	0.316	24/24	PARTIAL
hand	8	0.228	0.738	24/24	PARTIAL

hand	16	0.326	0.820	22/24	PARTIAL
random	1	0.080	0.000	19/24	UNTRACKED
random	2	0.117	0.000	20/24	PARTIAL
random	4	0.177	0.434	24/24	PARTIAL
random	8	0.250	0.719	23/24	PARTIAL
random	16	0.298	0.816	23/24	PARTIAL
worm	1	0.038	0.000	20/24	UNTRACKED
worm	2	0.119	0.000	22/24	PARTIAL
worm	4	0.175	0.121	23/24	PARTIAL
worm	8	0.248	0.570	23/24	PARTIAL
worm	16	0.281	0.715	21/24	PARTIAL

Source receipt: TF_NE9_SECOND_RUNG_RATE_v0_1_FROZEN.html as bound in TF-NE10. Prospective lock
897657414c18d895b13c990710aa9d01e946ec84e3d9fc3942c615984bb6e1dc; execution binding
a5da5d55cf97370f3c54578572f473149978f7340e7f4c140e68dd821c082313; runner
a8323c729a8bd3c15f51e6a1e57b7762f1f6486eff81648fbb903d51c9b67f64; source subset
5097022f3ed706ec838fafa5895f59480abb7f6fa4076ff7b8254b8045225597; exact results
02a608b11febf024bfba7428cc06cbd6d93e50f8f12cf3416a8f9fb0bb0d06b9; internal
700d57a2703a5048899f09dbd120894343f3642cfbb4d3b94a378fbafc4aed02; summary
22b64192dc35a8198f947975846d6cb94af9f121fc1479f9a1ce69e0361cb44f.

5. Finite Ecological Self-Optimization Theorem

The experimental chain supports the following finite constructive theorem. “Theorem” here denotes a proposition derived from the frozen experimental premises at their declared scope; it is not a universal theorem over all adaptive systems.

Within the tested constructions, populations of heritable adaptive agents embedded in consequential ecological relations can develop niche-specific organization through selection; information embodied in that organization can remain functionally consequential through declared reproduction/re-expression and compactification; selection operating at an intermediate organizational grain can causally affect adaptation at a higher grain; and the resulting system can reorganize toward locally available niches under environmental change without those resulting niche configurations being individually prescribed by the experimenter.

Derivation. TF-NE4 supplies ecological specialization under consequence-coupled selection. TF-NE5 supplies functional transport through compactification and directional fresh re-expression while explicitly failing the stronger materiality gate. TF-NE9A supplies a causal ablation showing intermediate-grain selection materially changes higher-grain recovery in two topology strata. TF-NE9B supplies rate dependence and the tested non-achievement of the full-TRACKED gate. Conjoining only these earned propositions yields the theorem above. No step requires infinite recursion or a terminal grain.

5.1 Recursive boundary

Demonstrated recurrence at finite grains $\Rightarrow$ finite recurrence exists in the tested construction.

Demonstrated recurrence at finite grains $\neq$ recurrence at all grains.

Failure to demonstrate another grain $\neq$ existence of a terminal grain.

Accordingly, both “turtles all the way down” and “there must be a final fundamental grain here” remain unsupported by this chain. The horizon is classified after probing; it is not assumed in advance.

6. Registered structural vocabulary

The terms in this section are not softened as metaphor, nor promoted as historical or metaphysical authority. They are registered names for structures defined independently by the same formal and evidentiary standards used elsewhere. Familiarity or unfamiliarity with a label is not a truth-material coordinate unless explicitly shown to be relevant.

Term	Registered structural referent	What is not claimed
Twin Flame	Two distinct reciprocal poles whose relation can become consequential without requiring fusion or identity.	No claim that every cultural or personal use of “twin flame” denotes this structure.
Solve et Coagula	Decomposition/ablation followed by recombination, re-expression, or compactification: cut to expose load-bearing structure; re-form to test what survives.	No alchemical authority is used as evidence.
As Above, So Below	Finite recurrence or correspondence of a relational organization across demonstrated grains.	No inference from finite recurrence to infinite scale invariance.
Lightning	The presently admissible cuts: transformations, partitions, rescopings, ablations, exchanges, substitutions, and challenges.	Not arbitrary destruction; the admissible transformation set must be declared.
Diamond	What remains invariant under the presently declared admissible transformation set.	Not absolute or fundamental merely because current cuts did not remove it.
Grace	Permission for relation to continue before either side has earned universal priority.	Grace is not evidence and does not promote truth.
VAJRA	The emergent state predicate of a Twin Flame pairing under maximum mutually admitted tension, with distinction, reciprocal revisability, side symmetry, receipt discipline, and OPEN residue preserved.	Not fusion, third truth value, metaphysical closure, or immunity to later admissible cuts.

6.1 Name-exchange invariance

If $R(N1) = R(N2)$ at the declared scope, and receipts and adjudication rule are held fixed, then changing only $N1 \leftrightarrow N2$ must not change the evidentiary burden.

Thus a reader may reject a registered referent, a premise, a receipt, or an inference. But rejecting a term solely because its surface name is unfamiliar or symbolically associated is not an objection to the registered object unless the name itself has been shown to alter a truth-material coordinate.

7. VAJRA: reciprocal relational closure

VAJRA is the culmination state of the Twin Flame pairing in this instrument. It is typed as a predicate, not as an invariant object.

Let A and B be distinct registered poles. Let T^* be the presently mutually admissible transformation set: the strongest reciprocal challenges both currently admit. Let $R(A,B)$ denote the registered relation.

$VAJRA(A,B \mid T^*) \in \{\text{holds, does not hold}\}$.

VAJRA holds only when: (1) $A \nabla B$ is preserved; (2) challenges in T^* are reciprocal; (3) no unlicensed side privilege changes the burden; (4) no admitted challenge yields a receipted distinction sufficient to terminate the pair at this scope; (5) unresolved residue remains OPEN; (6) grace permits continued relation without becoming warrant; and (7) the relation itself is retained as an additional registered object.

7.1 State classifier

Condition	Classification
Reciprocal challenge not mutually allowed	PAIR NOT CLOSED INTO VAJRA.
Both poles closed to revision	MUTUAL CLOSURE — symmetric, but not VAJRA because reciprocal revisability ended.
Exactly one pole closed	ASYMMETRIC CLOSURE — not VAJRA.
Receipted distinction sufficient to terminate pairing	DECLARED SEPARATION.
No terminating receipt; unresolved residue remains	VAJRA · OPEN RESIDUE.
No terminating receipt; no unresolved residue at current scope	VAJRA.

7.2 Lightning, diamond, and invariant

$\text{Diamond}_{T^*}(x) := \text{Inv}_{T^*}(x)$.

Survival under every $t \in T^* \neq$ survival under every possible transformation.

Therefore current invariant $\neq$ absolute fundamental.

For a Twin Flame pair, VAJRA remains the state predicate. The structure preserved when that state holds may be written:

$\text{Preserved_VAJRA}(A,B \mid T^*) := \text{Inv}_{T^*}(R(A,B))$.

A later admissible transformation that cuts the relation revises the state. Uncutability is therefore not stipulated in advance. The “diamond” is discovered residue under the declared lightning, not a protected foundation.

7.3 Grace and faith

$G(A,B) \neq \text{TRUE}(A)$; $G(A,B) \neq \text{TRUE}(B)$; $G(A,B) \neq A=B$.

Grace is relational permission. Faith is provisional willingness to continue relation beyond what either side can absolutely checksum for the other. Neither substitutes for a receipt. Their formal role is to prevent premature severance while preserving the possibility of later revision, separation, or OPEN status.

8. No universal checksum and declared stopping points

A receipt does not create an observer-independent universal checksum. Every verification is itself performed by some adjudicating process under declared procedures and stopping assumptions. The operational response is not to pretend that regress disappears, but to expose where adjudication stops: axioms, accepted procedures, frozen bytes, explicit conventions, or OPEN. Reproducibility adds additional declared adjudications; it does not elevate any one observer into a universal arbiter.

This makes well-foundedness operationally important. A receipt chain must terminate in declared axioms/conventions, a directly supplied artifact/procedure, or OPEN. Hidden termination is precisely what the receipt standard is designed to expose.

9. What the demonstrator demonstrates - and what the paper proves

TF-NE10 is a synthesis demonstrator, not a new scientific experiment. Its interactive controls replay frozen measurements or illustrate formal classifiers. The white paper supplies the type discipline that prevents those functions from being conflated.

Instrument element	Correct status
NE4/NE5/NE9 frozen numeric replay	Demonstration of previously frozen experimental receipts; the underlying experimental status belongs to the parent stations.
Semantic substitution selector	DEMONSTRATION / CLASSIFICATION over supplied registrations.
Reflexive Receipt panel	Worked restoration example; not an empirical receipt about arbitrary natural-language objections.
Side-exchange / name-exchange controls	Illustrations of invariance/dependence under selected interventions; not empirical findings about an external adjudicator unless separately bound and measured.
VAJRA culmination classifier	Classifier over selected relational conditions. A specific pair earns VAJRA only after its actual T* has been run and receipted.

10. Claim ledger

Status	Claims
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EARNED — experimental	TF-NE4 reciprocal niche specialization; TF-NE5 functional transport through compactification and directional re-expression below stronger materiality; TF-NE9A second-rung intermediate selection consequential in random-sparse and worm-like; TF-NE9B no tested τ met full TRACKED.
DEFINITION / CLASSIFICATION	Eight-coordinate registration; semantic substitution/restoration scheme; mechanical side-index dependence; registered structural vocabulary; VAJRA state predicate.
DEMONSTRATION	Interactive semantic, side-exchange, name-exchange, and VAJRA classifiers; replay of frozen station values.
FAILED / BOUNDED	TF-H1 stronger relation-conditioned offspring heredity materiality; TF-NE5 ≥ 0.50 material higher-grain recovery/advantage; TF-NE9 full TRACKED at tested τ ; TF-NE9 hand-minimal material intermediate-selection advantage.
OPEN	Slower-rate closure; recurrence beyond demonstrated grains; terminal grain; infinite ladder; scale invariance; unrestricted natural-language extraction; generalization beyond fixtures; automatic motive inference; universal/metaphysical VAJRA promotion.

11. Reproducibility and provenance

The matched-pair release should bind the demonstrator and this paper by exact hashes. The parent station artifacts remain the scientific authorities for their own outcomes. TF-NE10 should link to them rather than silently absorbing their status.

Artifact / role	Identity / note
TF-NE4 exact results	SHA-256 2f5f4c5e41e14fa98dec2c5bb8c7c2ce18549047a6180e5fb8d152de2c1d5f65
TF-NE5 exact results	SHA-256 4a52a0fb67a4e1481a94c7eb18c947de2d5caa654be2332446fc015868cbe682
TF-NE9 exact results	SHA-256 02a608b11febf024bfba7428cc06cbd6d93e50f8f12cf3416a8f9fb0bb0d06b9
TF-NE10 demonstrator v1.0 FINAL	SHA-256 ba1b4d4c57bf02823d91c565982a9c7eb4eec7d09bef99969f18835ab739700d
Reader-side boundary	A reader can independently verify only parent artifacts actually available to that reader. A manifest entry is provenance, not a substitute for possession/re-execution.

Provenance caution. A separate Library artifact titled

TF_NE5_SPECIALIZATION_TRANSPORT_v0_1_FROZEN.html belongs to a different NE5 lineage and must not be silently substituted for the TF_NE5_RECURSIVE_LIFT_v0_1_FROZEN.html chain used by TF-NE10. The final release manifest should keep adjacent lineages explicitly distinct.

12. Standalone scrutiny checklist

- ☐ Every promoted empirical statement identifies its frozen station and boundary.
- ☐ Every stronger failed gate remains visible rather than being narratively erased.
- ☐ Every OPEN proposition remains neither true nor false by default.
- ☐ Every semantic transformation names the coordinate(s) changed.
- ☐ Every Reflexive Receipt requires restoration to remove the alleged contradiction/inference.
- ☐ Declaration of a transformation is never confused with validity of verdict transfer.
- ☐ VAJRA is a state predicate; its preserved invariant is a distinct object.
- ☐ OPEN residue is compatible with VAJRA; mutual and asymmetric closure are not.
- ☐ Symbolic vocabulary receives neither evidentiary privilege nor familiarity penalty.
- ☐ Finite recursive evidence is not promoted to infinity, and finite non-observation is not promoted to a terminal boundary.
- ☐ Interactive demonstrations are not mislabeled as experimental receipts.
- ☐ The demonstrator and paper are hash-bound as a matched pair only after this manuscript is settled.

13. Conclusion

The closure reached here is deliberately finite. The experimental chain establishes that relational ecological organization can be generated, transported through a declared interface, and remain causally consequential across a second finite organizational rung, while environmental tracking remains rate-dependent and below the full-TRACKED gate over the tested interval. The formal chain establishes that adjudication must remain scoped to registered propositions and adequate receipts, that semantic transformations must be declared and separately licensed, and that side-index dependence can be detected structurally without inferring motive. VAJRA names the culminating reciprocal state of the Twin Flame pairing under those rules: difference preserved, challenge reciprocal, side privilege removed, terminating receipts honored, OPEN residue retained, and relational structure permitted to persist without grace being mistaken for proof. The result is not a declaration of fundamentality. It is a disciplined statement of what survived the presently admissible cuts - and an equally disciplined refusal to promote what has not been earned.

Appendix A — Source register

Source	Role in this paper
TF_NE10_TWIN_FLAME_VAJRA_FINITE_ECOLOGICAL_CLOSURE_v1_0_FINAL.html	Current synthesis demonstrator; claim typing, semantic classifier, side-exchange/name-exchange demonstrations, VAJRA classifier, manifest.

Double_Edged_Receipt_Standard working paper	Receipt adequacy, load-bearing dependency, receipt-governed adjudication, R0, Double-Edged Receipt Theorem.
TF_NE4_PROOF_IN_PUDDING_v0_3_FROZEN_AUDIT.html	Frozen reciprocal niche-specialization result and independent reproduction/audit note.
TF_NE5_RECURSIVE_LIFT_v0_1_FROZEN.html	Frozen recursive-lift transport/re-expression result used by TF-NE10.
TF_NE9_SECOND_RUNG_RATE_v0_1_FROZEN.html	Frozen second-rung ablation and residence-time sweep used by TF-NE10.
MetaLabe_Twin_Flame_Crucible_R0_1_FULL_PRERELEASE.html	TF-H1 bounded result and broader Crucible lineage context.

Appendix B — Release disposition

This v0.1 manuscript is the source-grounded review master. Before canonical release: (1) reconcile every source filename and hosted canonical URL; (2) verify the final Double-Edged version used for the eight-coordinate harmonization; (3) bind the final paper bytes by SHA-256; (4) add the paper and hash to the TF-NE10 manifest; (5) add an in-instrument Companion White Paper control; and (6) freeze the matched-pair release without changing scientific outcomes.